

(5)

$$T_n = \left\{ \psi: \mathbb{R} \rightarrow \mathbb{C}; \psi(x) = \sum_{k=-n}^n c_k e^{ikx}; c_k \in \mathbb{C} \right\}$$

$$e^{ikx} = \cos(kx) + i \sin(kx)$$

$$\langle \psi, \phi \rangle = \frac{1}{2\pi} \int_0^{2\pi} \psi(x) \overline{\phi(x)} dx$$

pokaži, da je $\Sigma = [e^{-inx}, e^{-i(n-1)x}, \dots, 1, e^{ix}, \dots, e^{inx}]$ ONB prostora T_n .

$$\begin{aligned} \frac{d}{dx}(e^{i\alpha x}) &= \frac{d}{dx}(\cos(\alpha x) + i \sin(\alpha x)) = -\alpha \sin(\alpha x) + i\alpha \cos(\alpha x) = \\ &= i\alpha(\sin(\alpha x) + i \cos(\alpha x)) = \underline{i\alpha e^{i\alpha x}} \end{aligned}$$

$$|e^{ikx}| = \langle e^{ikx}, e^{ikx} \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{ikx} \overline{e^{ikx}} dx = \frac{1}{2\pi} \int_0^{2\pi} e^{ikx} e^{-ikx} dx = \frac{1}{2\pi} \int_0^{2\pi} e^0 dx = 1$$

$$\langle e^{ikx}, e^{ilx} \rangle = \frac{1}{2\pi} \int_0^{2\pi} e^{ikx} \overline{e^{ilx}} dx = \frac{1}{2\pi} \int_0^{2\pi} e^{i(k-l)x} dx = \frac{1}{2\pi i(k-l)} e^{i(k-l)x} \Big|_0^{2\pi} =$$

$$= \frac{1}{2\pi i(k-l)} \left(\underbrace{e^{i(k-l)2\pi} - 1}_{=0} \right) = 0$$

$\Rightarrow \Sigma$ je ortonormiran sistem

zaradi definicije $T_n \Rightarrow \Sigma = \text{ONB}$

① Izračunaj adj. end.:

a) $W: \mathbb{R}^3 \rightarrow \mathbb{R}^3$; $W(x) = \vec{w} \times \vec{x}$; $\vec{w} \in \mathbb{R}^3$

$$\varepsilon = [\vec{i}, \vec{j}, \vec{k}]$$

$$[W]_{\varepsilon}^{\varepsilon} = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix}$$

$$[W^*]_{\varepsilon}^{\varepsilon} = \begin{bmatrix} 0 & w_z & -w_y \\ -w_z & 0 & w_x \\ w_y & -w_x & 0 \end{bmatrix} = -[W]_{\varepsilon}^{\varepsilon}$$

$$W(\vec{i}) = \vec{w} \times \vec{i} = \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ \vec{w}_x & \vec{w}_y & \vec{w}_z \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ w_z \\ -w_y \end{bmatrix}$$

$$W(\vec{j}) = \vec{w} \times \vec{j} = \begin{bmatrix} -w_z \\ 0 \\ w_x \end{bmatrix}$$

$$W(\vec{k}) = \vec{w} \times \vec{k} = \begin{bmatrix} w_y \\ -w_x \\ 0 \end{bmatrix}$$

2. NAČIN

$$\begin{aligned} \langle \vec{x}, W^* \vec{y} \rangle &= \langle W \vec{x}, \vec{y} \rangle = (\vec{w} \times \vec{x}) \cdot \vec{y} = (\vec{y} \times \vec{w}) \cdot \vec{x} = -(\vec{w} \times \vec{y}) \cdot \vec{x} = \langle \vec{x}, -(\vec{w} \times \vec{y}) \rangle = \\ &= \langle \vec{x}, -W \vec{y} \rangle \Rightarrow \underline{\underline{W^* = -W}} \end{aligned}$$

b) T_n

$$D: T_n \rightarrow T_n$$

$$D(\psi) = \psi'$$

$$D^* = ?$$

$$\varepsilon = [e^{-inx}, \dots, 1, \dots, e^{inx}]$$

$$D(e^{ikx}) = (e^{ikx})' = ike^{ikx}$$

$$[D]_{\varepsilon}^{\varepsilon} = \text{diag}(-in, -i(n-1), \dots, 0, \dots, in)$$

$$[D^*]_{\varepsilon}^{\varepsilon} = \text{diag}(-in, \dots, 0, \dots, in)^h =$$

$$= \underline{\underline{\text{diag}(in, \dots, 0, \dots, -in)}} = -[D]_{\varepsilon}^{\varepsilon} = [-D]_{\varepsilon}^{\varepsilon}$$

$$\underline{\underline{D^* = -D}}$$

$\mathbb{R}^n$

Definiramo

$$\langle u, v \rangle_g = \sum_{i,j=1}^n a_{ij} u_i v_j \quad ; \quad a_{ij} = a_{ji}$$

$$g = [a_{ij}]_{i,j=1,\dots,n} \in \text{Mat}(n \times n, \mathbb{R})$$

$$= \sum_{i=1}^n v_i \sum_{j=1}^n a_{ij} u_j = \sum_{i=1}^n v_i (gu)_i = \boxed{v^t (gu)} = \langle u, v \rangle_g$$

$$\downarrow$$

$$\langle u, v \rangle_g = \langle v, u \rangle_g$$

$$\langle v, v \rangle_g = v^t g v \geq 0$$

$$\parallel$$

$$\langle g v, v \rangle$$

$$\boxed{g \text{ pozitivno definitna}}$$

$$A \in \text{Mat}(n \times n, \mathbb{R})$$

$$A_g^* \text{ adj. presl. glede na } \langle \cdot, \cdot \rangle_g$$

$$v^t (gu) = u^t (gv)$$

$$\downarrow$$

$$v^t g^t u$$

$$g = g^t$$

$$\langle u, A_g^* v \rangle = (A_g^* v)^t g u =$$

$$= v^t (A_g^*)^t g u =$$

$$= \langle A u, v \rangle_g \rightarrow (A_g^*)^t g = g A$$

$$= v^t g A u$$

$$(A_g^*)^t = g A g^{-1}$$

$$A_g^* = (g^{-1})^t A^t g^t = \underline{\underline{g^{-1} A^t g}}$$

(2.) $\mathbb{R}^2$

$$\langle x, y \rangle_g = 2x_1 y_1 + 2x_2 y_2 + x_1 y_2 + x_2 y_1$$

$$g = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$A: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$A(t, s) = (t, 2s)$$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A_g^* = g^{-1} A^t g = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \underline{\underline{\frac{1}{3} \begin{bmatrix} 2 & -2 \\ 2 & 7 \end{bmatrix}}}$$

$$(3) \quad A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 2 \end{bmatrix}$$

$$A = P D P^t$$

$$P P^t = I$$

$$p_A(\lambda) = (2-\lambda)(1-\lambda)(2-\lambda) - (1-\lambda) = (1-\lambda)((2-\lambda)^2 - 1) = (1-\lambda)(2-\lambda+1)(2-\lambda-1) = (1-\lambda)^2(3-\lambda)$$

$$\lambda_1 = 1 \quad \lambda_2 = 3$$

$$\dim \ker(1) = 2$$

ker A symmetrisch, j. diag. $\Rightarrow$

$$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

$$\lambda = 1:$$

$$(A - I)_V = 0$$

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} x+z &= 0 \\ x &= -z \end{aligned}$$

$$r = y(0, 1, 0) + z(-1, 0, 1)$$

$$E_1(1) = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\} = \text{span} \left\{ \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -\frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

$$\lambda = 3:$$

$$(A - 3I)_V = 0$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -2 & 0 \\ 1 & 0 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} y &= 0 \\ x &= z \end{aligned}$$

$$\vec{r} = z(1, 0, 1)$$

$$E_1(3) = \text{span} \left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\}$$

$$P = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$$